CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 2

Chapter 1: **Chemical Reactions and Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$ is a: 1

(a) decomposition reaction **(b)** combination reaction
(c) displacement reaction **(d)** double displacement reaction

2. When an iron nail is placed in copper sulphate solution, the deposit on the nail is: 1

(a) reddish-brown **(b)** black **(c)** green **(d)** white

3. The symbol (aq) in an equation means that the substance is: 1

(a) dissolved in water **(b)** a gas
(c) a solid **(d)** a liquid

4. A reaction in which heat is given out is called: 1

(a) displacement **(b)** exothermic **(c)** endothermic **(d)** decomposition

5. Rancidity occurs in: 1

(a) metals **(b)** glass **(c)** water **(d)** fats and oils

6. The symbol (g) in a chemical equation represents: 1

(a) solid state **(b)** liquid state **(c)** aqueous solution **(d)** gaseous state

7. A reaction in which heat is absorbed is called: 1

(a) exothermic **(b)** endothermic **(c)** precipitation **(d)** combination

8. Rusting of iron requires: 1

(a) air and moisture **(b)** only sunlight **(c)** only dry air **(d)** only nitrogen

9. Which of the following is a chemical change? 1

(a) boiling of water **(b)** melting of ice
(c) dissolving sugar in water **(d)** souring of milk

10. Burning of natural gas is: 1

(a) exothermic **(b)** a displacement reaction
(c) neither exo nor endo **(d)** endothermic

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Define a displacement reaction and give an example. 2

12. What is rancidity? 2

13. What is a balanced chemical equation? 2

14. What is a precipitate? Give one example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Balance the equation $\mathrm{Mg} + \mathrm{O_2} \longrightarrow \mathrm{MgO}$. 3

16. Name the type of reaction: $\mathrm{Zn} + \mathrm{H_2SO_4} \longrightarrow \mathrm{ZnSO_4} + \mathrm{H_2}$. Give a reason. 3

17. Give three examples of exothermic reactions. 3

OR

Balance the equation $\mathrm{H_2} + \mathrm{O_2} \longrightarrow \mathrm{H_2O}$. 3

18. Name the type of reaction: $\mathrm{AgNO_3} + \mathrm{NaCl} \longrightarrow \mathrm{AgCl} + \mathrm{NaNO_3}$. Give the colour of the precipitate. 3

OR

Give three examples of endothermic reactions. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Zinc granules are added to dilute sulphuric acid. Write the observation, balanced equation, type, and the test for the gas. 5

OR

Differentiate between exothermic and endothermic reactions with two examples of each. 5

20. Electric current is passed through water. Name the gases at the two electrodes, give the equation and the volume ratio. 5

OR

Quick lime is added to water in a beaker. State the observation, write the equation, name the product and give one use. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND